A set closed under algebraic closure is an elementary subsystem

Silvia Barbina, Enrique Casanovas

arxiv:2411.13723v3; Creative Commons Attribution 4.0 International

Editorial note. The counted statement is the author's environment at source0.tex lines 789-790 and its proof environment opens at line 791; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

Editorial note. Metadata provenance: the source record of the pinned copy and the licence evidence file are both present; the licence is CC BY 4.0 with status supported, and the card's source_version matches the value recorded in that evidence file.

Editorial note. r002 note: r001 failed because the card's own materials prose, which the wrapper typesets, carried characters the font does not provide; r002 replaces them by equivalent TeX commands and changes nothing else, and no retained author line is altered.

Editorial note. r003 note: the retained author proof cites the bibliography keys pmneumann and hodges, so r003 bundles the article's own bibliography file; r002 had been built with the wrong flag and changed nothing.

Editorial note. r004 note: the retained proof cites the bibliography keys pmneumann and hodges, which the prepared per-source bibliography of this source does not carry; r004 adds a task-directory bibliography.tex holding those two entries copied character for character from the article's own thebibliography

block and points bibliography_paths at it. No bibliographic datum is invented and no retained author line is altered.

Editorial scope. One original result of the pinned author manuscript, counted once: a set equal to its algebraic closure is an elementary subsystem (author label Feb_23_2024_8).

Allowed standard prerequisites (editorial).

- Algebraic closure. The algebraic closure of A is the set of elements whose type over A is algebraic, and it is itself closed under the closure.. It is the hypothesis A equals its algebraic closure.
- Unconfined system. A Steiner triple system admitting an HF-ordering without maximum.. It is the ambient hypothesis.
- Neumann's lemma. Given elements outside the algebraic closure of A, any set in the monster model can be moved so that the chosen elements avoid it, over A.. It supplies the back-and-forth move used in the proof.

Lemma 0.1 If $<$ is an HF-ordering of B and $A = \mathrm{cl}_<(A) \subseteq B$, then $<$ is an HF-ordering of B over A.

Proposition 0.2 If M is an unconfined STS and $A = \mathrm{acl}(A) \subseteq M$, then $A \leq M$.

Proof. We can expand the monster model $\mathbb{M}$ by adding an HF-ordering $<$ of $\mathbb{M}$, but we work in $\mathbb{M}$. By P.M. Neumann's Lemma (2.3 in [1] — also [2, Corollary 4.2.2]), given $a_1, \ldots, a_n \notin \mathrm{acl}(A)$, for any set B in $\mathbb{M}$ there are $b_1 \ldots b_n$ with $b_1, \ldots, b_n \equiv_A a_1 \ldots a_n$ such that $b_1, \ldots, b_n \notin B$. Hence if $M \setminus A = \{a_i \mid i < \kappa\}$ and $p(x_i \mid i < \kappa) = \mathrm{tp}((a_i \mid i < \kappa)/A)$, then, for any set B, the type p has a realization $(b_i \mid i < \kappa)$ such that $\{b_i \mid i < \kappa\} \cap B = \emptyset$. Using this, for any λ we can construct a sequence of models $(M_i \mid i < \lambda)$ containing A such that $M = M_0$, $M_i \cong_A M_j$ and $M_i \cap M_j = A$ for every $i \neq j$. Let $B = \mathrm{cl}_<(A) \subseteq \mathbb{M}$ and let $\mu = |B| = |A| + \omega$. We may assume $\mu < \lambda$, which implies $M_i \cap B = A$ for some i (if $A \subsetneq M_i \cap B$ for every $i < \lambda$, we can choose $a_i \in (M_i \cap B) \setminus A$ and the a_i are all distinct, which contradicts $|B| < \lambda$). By Lemma 0.1, $<$ is an HF-ordering over B. Therefore $<_i = <\restriction M_i$ is an HF-ordering of M_i over $A = B \cap M_i$. Since $M_i \cong_A M$, there is an HF-ordering of M over A. □

References

[1] P.M. Neumann. The structure of finitary permutation groups. Archiv der Mathematik, 27:3–17, 1976.

[2] W. Hodges. Model theory. Cambridge University Press, 1993.